

VDMM-BESP Users Guide

Version 1.0

Contents

1. Introduction	1
2. Installation and Startup.....	1
2.1 Environment Requirements	1
2.2 Launch MATLAB via COMSOL.....	1
2.3 Software Launch	2
2.3.1 Launch via Source Code (Recommended).....	2
2.3.2 Launch via Packaged App	4
2.4 Database Configuration (Optional).....	4
2.4.1 Create Database and Tables.....	4
2.4.2 Configure ODBC Data Source	5
3. Usage Guide	6
3.1 Single Model Calculation	6
3.1.1 Parameter Definitions.....	6
3.1.2 Electrode Matrix Configuration	7
3.1.3 Running Simulation.....	7
3.2 Batch Model Calculation	8
3.2.1 Export Options Configuration.....	8
3.2.2 Calculation Modes.....	8
3.2.3 Batch Parameter Input Format	9
3.2.4 Running Batch Calculation	9
3.3 Data Analysis	9
3.3.1 Data Loading and Selection	10
3.3.2 Selection of Data Analysis Methods	10
3.4 Prediction Model Construction.....	13
3.4.1 Data Preparation and Loading.....	13
3.4.2 Model Selection and Hyperparameters	13
3.4.3 Training and Evaluation	14
3.5 Model Prediction (Optional).....	15

1. Introduction

VDMM-BESP is a batch simulation tool developed using MATLAB App Designer. Integrated with the COMSOL Multiphysics finite element simulation engine, this software is designed to simulate and analyze the potential distribution characteristics of metal pipelines with pitting defects under electric field excitation. Combined with data analysis capabilities, it provides a complete workflow covering physical modeling, batch sample generation, statistical analysis, and predictive modeling.

2. Installation and Startup

2.1 Environment Requirements

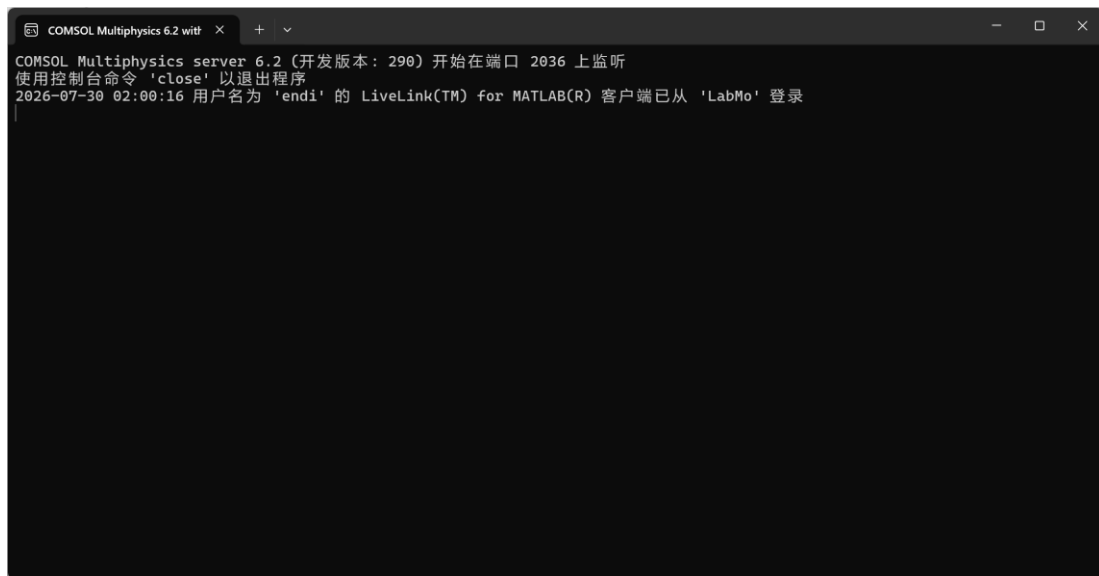
The hardware and software environment requirements for running the program are specified in the table below:

Item	Requirement	Remark
Operating System	Windows 10/11 (64-bit)	Windows 11 is recommended
MATLAB	R2022a or later	App Designer component is required
COMSOL Multiphysics	COMSOL Multiphysics 6.2 with LiveLink for MATLAB	For finite element electric field simulation calculation
Database (Optional)	MySQL 8.0+ and ODBC Driver	For long-term storage and multi-user shared querying of simulation results.

2.2 Launch MATLAB via COMSOL

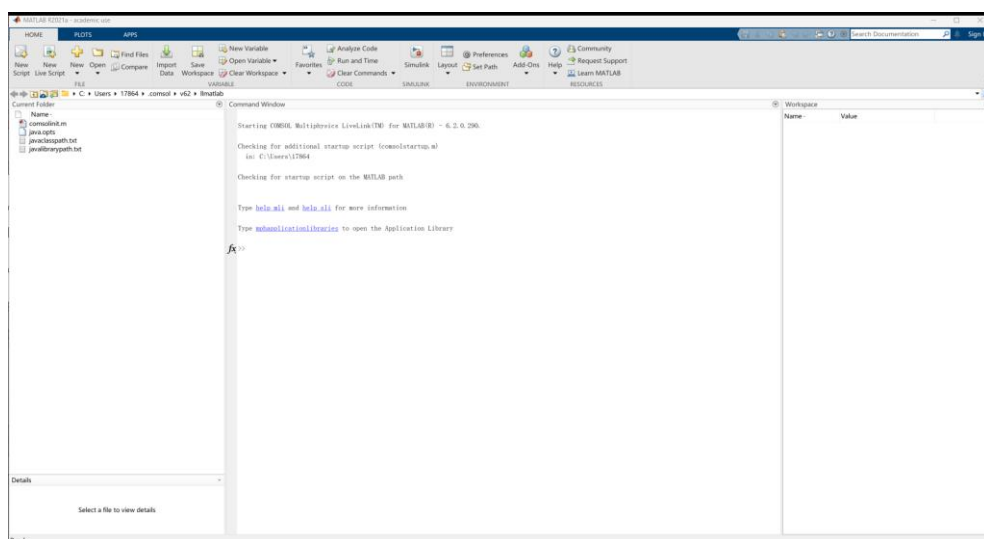
If COMSOL Multiphysics 6.2 with LiveLink for MATLAB is installed, follow the steps below to launch the MATLAB environment:

Step 1: Double-click the desktop icon of "COMSOL Multiphysics 6.2 with MATLAB" to open the command line terminal. A username and password are required for the first launch, and no re-verification is needed for subsequent use.



Step 2: COMSOL will automatically launch and load the MATLAB environment. Wait until the MATLAB main interface (including the Command Window, Workspace, and toolbar) is fully loaded.

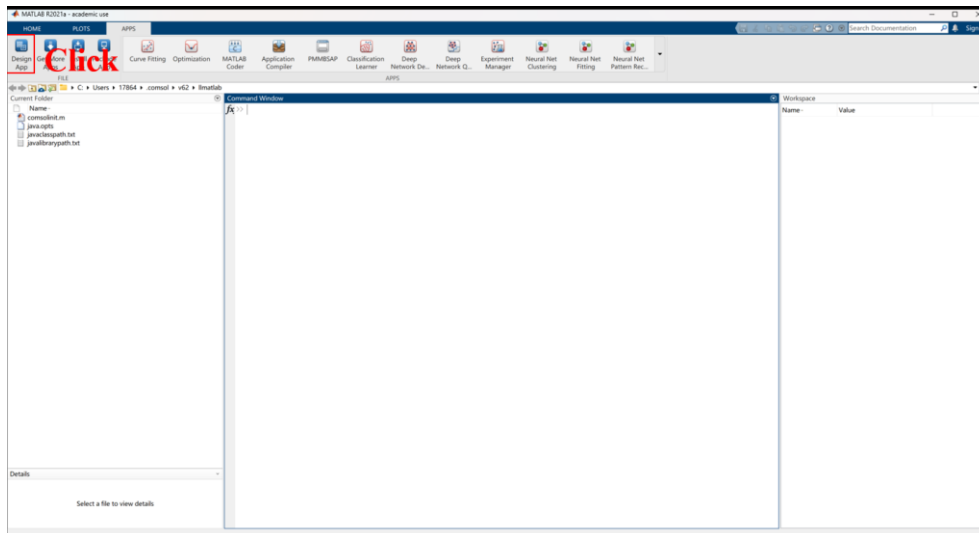
Step 3: Confirm that the COMSOL LiveLink initialization success message appears in the MATLAB command window.



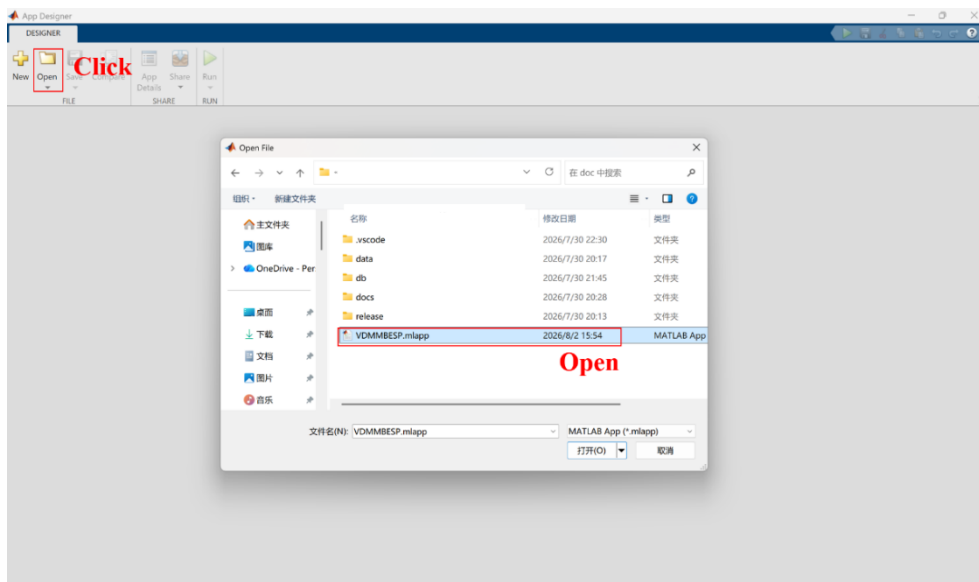
2.3 Software Launch

2.3.1 Launch via Source Code (Recommended)

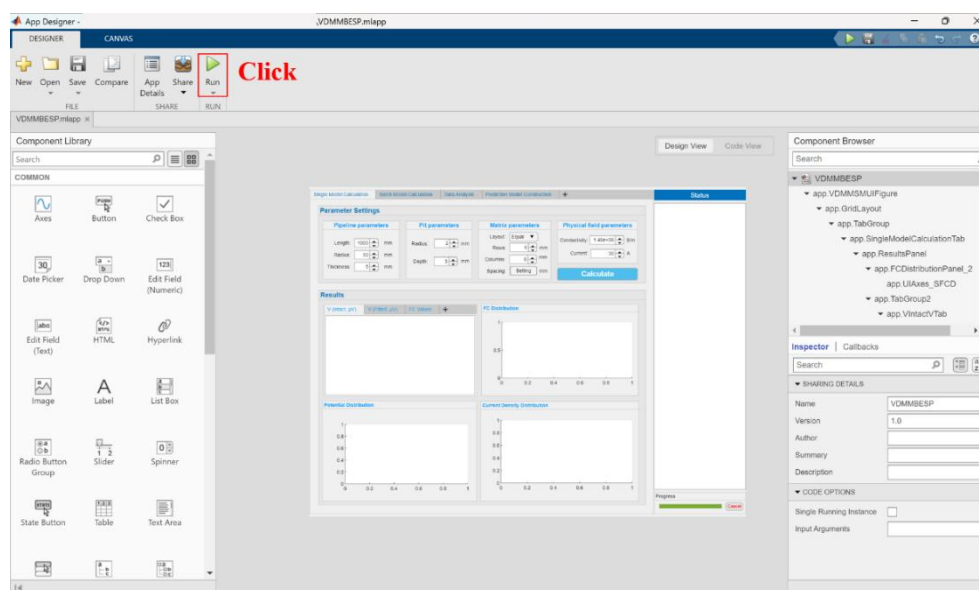
Step 1: Open MATLAB App Designer.



Step 2: Import and open the VDDMM-BESP.mlapp file.

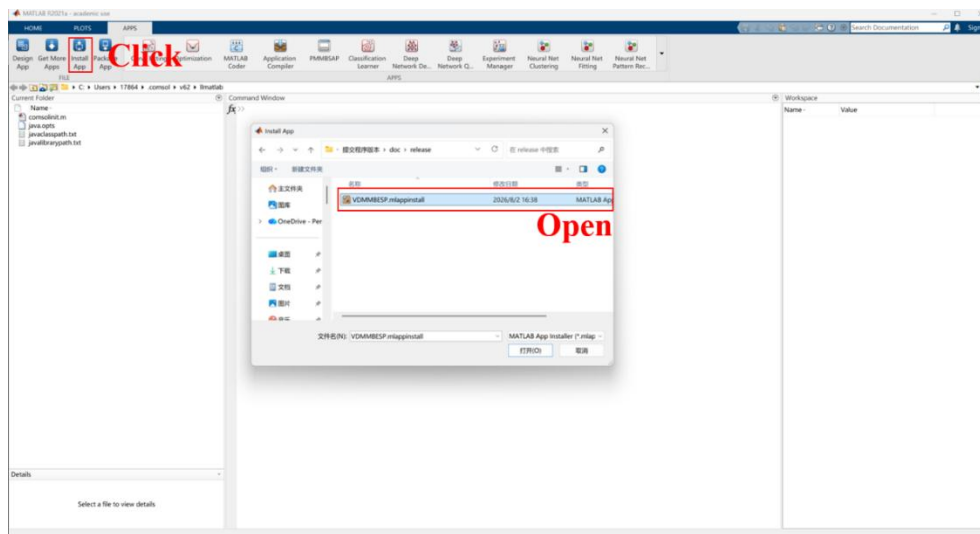


Step 3: Click the Run button to start the software.

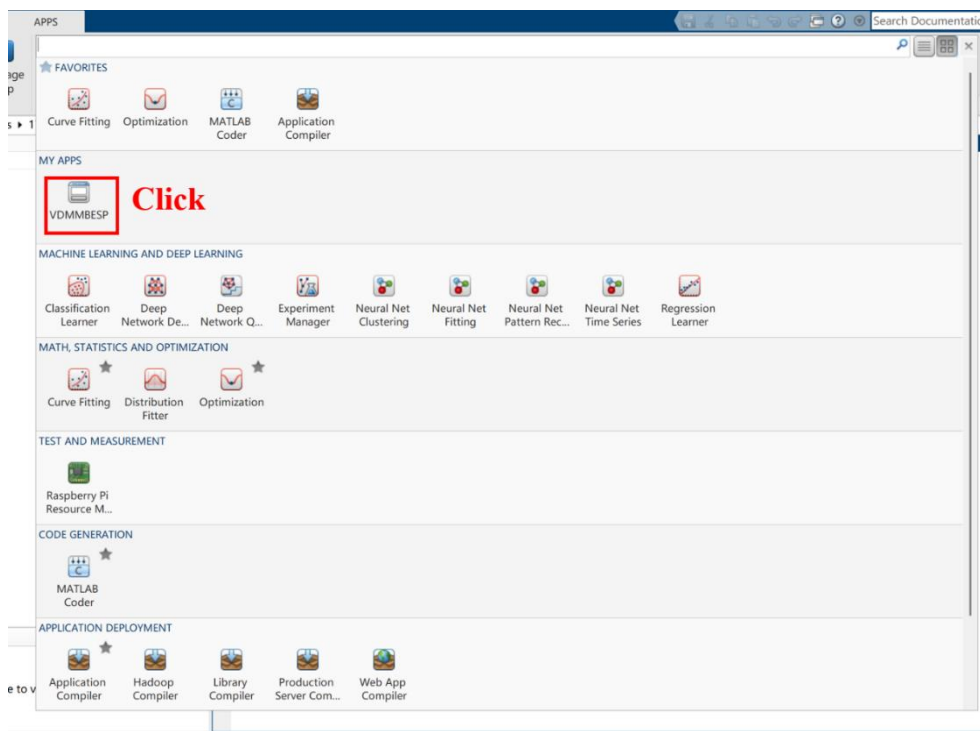


2.3.2 Launch via Packaged App

Step 1: Click **Install App** and import the "release/VDMM-BESP.mlappinstall" file.



Step 2: After installation, locate the VDMM-BESP icon in the **APPS** list and click it to run the application.



2.4 Database Configuration (Optional)

Configure the database if you need to store batch calculation results in MySQL for long-term storage and multi-user shared querying. Skip this section if you only need the Excel and MAT file export functions.

2.4.1 Create Database and Tables

Open the Windows Terminal, connect to the MySQL database, and create the corresponding

database and data tables. The SQL file for table creation is provided in `..\db\simulation_records.sql`.

```
C:\WINDOWS\system32\cmd.exe
Microsoft Windows [版本 10.0.26200.8875]
(c) Microsoft Corporation. 保留所有权利。

C:\Users\17864>mysql -u root -p 1. Login to the database
Enter password: *****
Welcome to the MySQL monitor.  Commands end with ; or \g.
Your MySQL connection id is 47
Server version: 8.0.26 MySQL Community Server - GPL

Copyright (c) 2000, 2021, Oracle and/or its affiliates.

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affiliates. Other names may be trademarks of their respective
owners.

Type 'help;' or '\h' for help. Type '\c' to clear the current input statement.

mysql> CREATE DATABASE IF NOT EXISTS pipepitssim CHARACTER SET utf8mb4 COLLATE utf8mb4_unicode_ci; 2. Create a database
Query OK, 1 row affected (0.00 sec)

mysql> use pipepitssim; 3. Using the database
Database changed
mysql> source oc\db\simulation_records.sql 4. Create table structure
Query OK, 0 rows affected (0.00 sec)

Query OK, 0 rows affected (0.00 sec)

Query OK, 0 rows affected (0.00 sec)

Query OK, 0 rows affected (0.00 sec)

Query OK, 0 rows affected (0.00 sec)
```

2.4.2 Configure ODBC Data Source

Open the Windows Terminal, execute the command `C:\Windows\SysWOW64\odbcad32.exe` to launch the ODBC Data Source Administrator, and add a new data source following the graphical instructions.



3. Usage Guide

The graphical interface consists of four tabs, corresponding to different stages of the simulation and analysis workflow:

Single Model Calculation: Perform dual-model simulation with a complete set of fixed parameters to view the voltage matrix, *FC* distribution, electric potential distribution, and current density field.

Batch Model Calculation: Automatically scans multiple parameter sets, generates a large number of simulation samples in batches, and exports the corresponding data.

Data Analysis: Load data from multiple sources, filter data by parameters, visualize single-model *FC* results, and conduct correlation analysis and regression fitting.

Prediction Model Construction: Build prediction models (BP, RF, GBDT, and SER) to predict pitting parameters from *FC* values, and output the model evaluation metrics.

3.1 Single Model Calculation

This module is used to conduct a complete electric field simulation calculation for a set of fixed parameter values.

3.1.1 Parameter Definitions

The definitions of relevant simulation parameters are shown in the table below:

Type	Parameter Name	Abbreviation	Unit	Description
Pipeline parameters	Length	PipL	mm	Pipeline length
	Radius	PipR	mm	Pipeline radius
	Thickness	PipT	mm	Pipeline wall thickness
Pit parameters	Radius	Pitr	mm	Pitting radius
	Depth	Pith	mm	Pitting depth
Matrix parameters	Rows	Rows	/	Row count of electrode matrix
	Columns	Cols	/	Column count of electrode matrix
	Spacing	Spacing	mm	Electrode matrix spacing
Physical field parameters	Conductivity	/	S/m	Electrical conductivity of material
	Current	/	A	Excitation current value

3.1.2 Electrode Matrix Configuration

VDMM-BESP supports two electrode deployment schemes, which can be switched via the **Layout** dropdown box:

Equal: After setting the Rows, Columns, and Spacing parameters, the software automatically calculates the three-dimensional coordinates of each electrode on the pipeline surface.

Customize: Click the **Settings** button to open the **Electrode Coordinate Table** dialog. Users can manually specify the three-dimensional coordinates (x, y, z) of each electrode or import coordinate data from CSV files.

The image shows three variations of the 'Matrix parameters' dialog box and one 'Electrode Coordinate Table' dialog box.

The first 'Matrix parameters' dialog box has 'Layout' set to 'Equal', 'Rows' set to 'Equal', 'Columns' set to 'Customize', and 'Spacing' set to 30 mm.

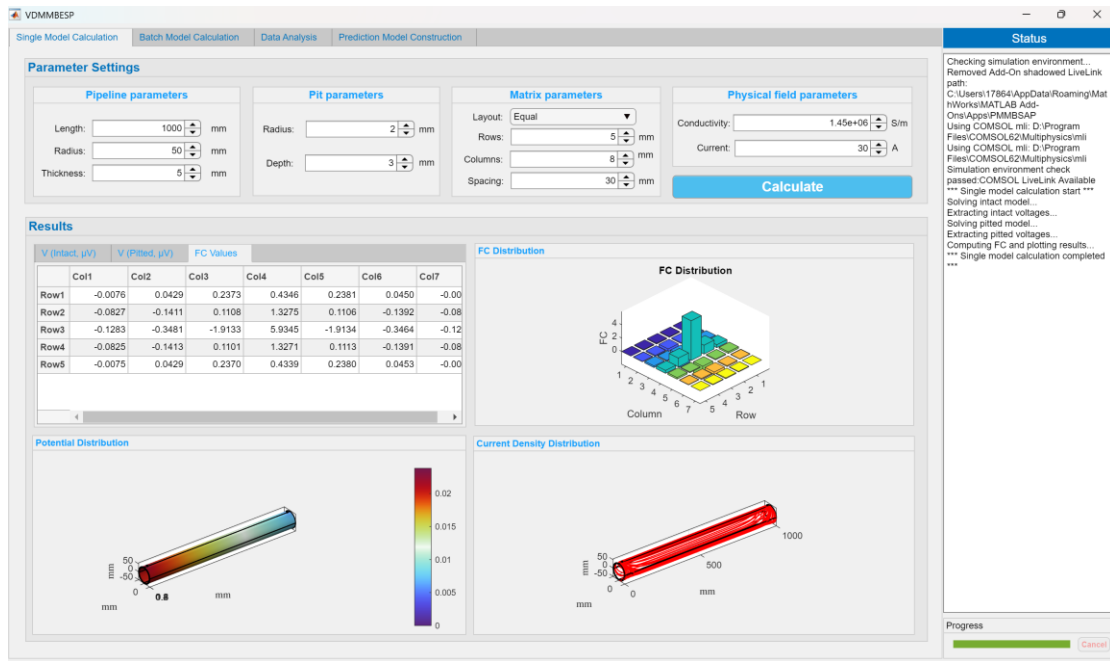
The second 'Matrix parameters' dialog box has 'Layout' set to 'Equal', 'Rows' set to 5 mm, 'Columns' set to 8 mm, and 'Spacing' set to 30 mm.

The third 'Matrix parameters' dialog box has 'Layout' set to 'Customize', 'Rows' set to 5 mm, 'Columns' set to 8 mm, and a 'Setting' button.

The 'Electrode Coordinate Table' dialog box is a window with a table containing 10 columns (Column 1 to Column 10) and 4 rows (x, y, z, and an empty row). The table is currently empty. Below the table are three buttons: 'Load CSV File', 'Export to CSV', and 'Save to App Variable'.

3.1.3 Running Simulation

Click the **Calculate** button to execute the electric field distribution simulation under the currently set parameters. The output results include the voltage values of non-corroded and pitting-corroded pipelines, the *FC* values and distribution of pitting-corroded pipelines, and the cloud maps of pipeline potential and current density distribution.



3.2 Batch Model Calculation

The batch calculation module supports automatic repeated simulation within the parameter space to generate large volumes of sample data efficiently.

3.2.1 Export Options Configuration

Configure export parameters and file save paths in the **Export Options** and **File Save Path** panels before starting batch calculation:

Checkbox Option	Description	Remark
Export to Excel	Save parameters and <i>FC</i> results as .xlsx table files	Require a customized save path and file name
Export to MAT-file	Save complete calculation results as .mat files	Require customized save path and file name
Save Report	Generate text reports with titles, timestamps, and operation logs	Require customized save path and report name
Save to Database	Write simulation data to MySQL <code>simulation_records</code> table	Valid only after successful database connection
Select All	Enable all export functions with one click	/

3.2.2 Calculation Modes

Full Scan: Perform Cartesian product traversal and simulation for all parameter values.

Combination: Conduct calculation based on one-to-one matching of pipeline radius and wall

thickness. When the **Enable Scan** checkbox is checked, full traversal calculation of pitting parameters will also be performed.

3.2.3 Batch Parameter Input Format

The batch parameter input box supports multiple input formats for multi-group parameter values, as shown in the table below:

Input Type	Input Content	Calculated Parameter Values
Space-separated	1 2 3 4	[1, 2, 3, 4]
Comma-separated	1, 2, 3, 4	[1, 2, 3, 4]
Colon expression	1:1:4	[1, 2, 3, 4]
Mixed format	1, 2:1:4	[1, 2, 3, 4]

3.2.4 Running Batch Calculation

Click the **Calculate** button to start batch calculation. You can view the real-time progress percentage, elapsed time, and operation logs during the calculation process. After the batch calculation is completed, the **Results** panel provides two levels of data viewing functions:

Summary View: Displays brief information of all calculation records.

Detailed View: Shows complete, detailed data for batch calculation results.

Result Saving

Export Options

- ☒ Select All
- ☒ Export to Excel
- ☒ Export to MAT-file
- ☒ Save Report
- ☒ Save to Database

File Save Path

Path: D:\Data

File Name: results01

Report Name: report01

Database Configuration

Data Source: test01

Username: root

Password: 123456

Status: ●

Batch Calculation Parameters

Calculation Mode

- ☒ Full Scan
- ☐ Combination

Pipeline parameters

Length: 1000 mm

Radius: 60 mm

Thickness: 6 mm

Pits parameters

Radius: 1.5/0.1/5 mm

Depth: 1.5/0.1/5 mm

☐ Enable Scan

Matrix parameters

Layout: Equal

Rows: 3

Columns: 4

Spacing: 15.5/30 mm

Results

Total Models	Calculation Time (s)	Max Voltage (Intact)	Max Voltage (Pit)	Max FC	Min FC
5184	1.0575e+04	289.0666	304.0213	186.8666	-58.5246

Index	PipeL	PipeR	PipeS	pitr	pith	rows	cols	spacing	sigma	current	FC_1	FC_2	FC_3	FC_4
1	1000	60	6	1.5000	1.5000	3	4	15	1450000	30	0.0638	1.0952	0.0606	-1.4015
2	1000	60	6	1.5000	1.5000	3	4	20	1450000	30	0.0385	0.6217	0.0365	-0.8872
3	1000	60	6	1.5000	1.5000	3	4	25	1450000	30	0.0275	0.4008	0.0260	-0.5841
4	1000	60	6	1.5000	1.5000	3	4	30	1450000	30	0.0215	0.2810	0.0200	-0.4084
5	1000	60	6	1.5000	1.6000	3	4	15	1450000	30	0.0674	1.1847	0.0645	-1.5191
6	1000	60	6	1.5000	1.6000	3	4	20	1450000	30	0.0409	0.6711	0.0398	-0.9614

Status

time=9996.6s, progress=98.3%
 [1281/1296] L=1000 R=60 S=6 r=5
 h=3.5
 time=10031.0s, progress=98.8%
 [1282/1296] L=1000 R=60 S=6 r=5
 h=3.6
 time=10066.9s, progress=99.0%
 [1283/1296] L=1000 R=60 S=6 r=5
 h=3.7
 time=10100.5s, progress=99.0%
 [1284/1296] L=1000 R=60 S=6 r=5
 h=3.8
 time=10135.6s, progress=99.1%
 [1285/1296] L=1000 R=60 S=6 r=5
 h=3.9
 time=10171.6s, progress=99.2%
 [1286/1296] L=1000 R=60 S=6 r=5
 h=4
 time=10207.6s, progress=99.2%
 [1287/1296] L=1000 R=60 S=6 r=5
 h=4.1
 time=10242.2s, progress=99.3%
 [1288/1296] L=1000 R=60 S=6 r=5
 h=4.2
 time=10279.1s, progress=99.4%
 [1289/1296] L=1000 R=60 S=6 r=5
 h=4.3
 time=10316.0s, progress=99.5%
 [1290/1296] L=1000 R=60 S=6 r=5
 h=4.4
 time=10354.0s, progress=99.5%
 [1291/1296] L=1000 R=60 S=6 r=5
 h=4.5
 time=10394.4s, progress=99.6%
 [1292/1296] L=1000 R=60 S=6 r=5
 h=4.6
 time=10429.1s, progress=99.7%
 [1293/1296] L=1000 R=60 S=6 r=5
 h=4.7
 time=10467.3s, progress=99.8%
 [1294/1296] L=1000 R=60 S=6 r=5
 h=4.8
 time=10503.5s, progress=99.8%
 [1295/1296] L=1000 R=60 S=6 r=5
 h=4.9
 time=10541.1s, progress=99.9%
 [1296/1296] L=1000 R=60 S=6 r=5
 h=5
 time=10577.5s, progress=100.0%
 Batch finished: 5184 rows in ResultsTable, elapsed=10577.5s

Progress:

3.3 Data Analysis

The data analysis tab provides a complete statistical analysis workflow, including data loading and screening, correlation analysis, univariate regression, and multivariate regression modeling.

3.3.1 Data Loading and Selection

Select the data source type via the **Source** dropdown box in the **Data Selection** panel:

Type	Description
Batch	Read results of batch calculation in the current session
Excel	Import data from .xlsx files
MAT	Import data from .mat files
Database	Query MySQL data by parameter range (database connection required)

Click the **Load** button to open the data filtering dialog. Users can filter batch calculation records according to custom conditions, and select to display all data or specified row ranges via the **Data Range** dropdown box. Click **Save** to confirm and store the filtered data.

Select Data (Excel)

Pipeline (min ~ max)

Length (L): 1200 ~ 1200

Radius (R): 50 ~ 90

Thickness (S): 5 ~ 7

Pitting (min ~ max)

Radius (r): 2 ~ 4

Depth (h): 2 ~ 4

Electrode

Rows: 3 ~ Cols: 4

Spacing: 15 ~ 40

Physical Field (single value)

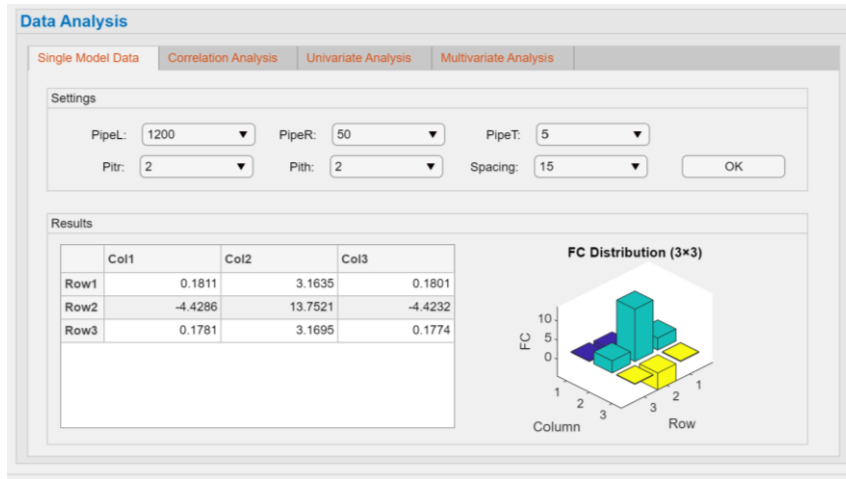
Sigma: 1450000 ~ Current: 30

Confirm Cancel

3.3.2 Selection of Data Analysis Methods

(1) Single Variable Data View

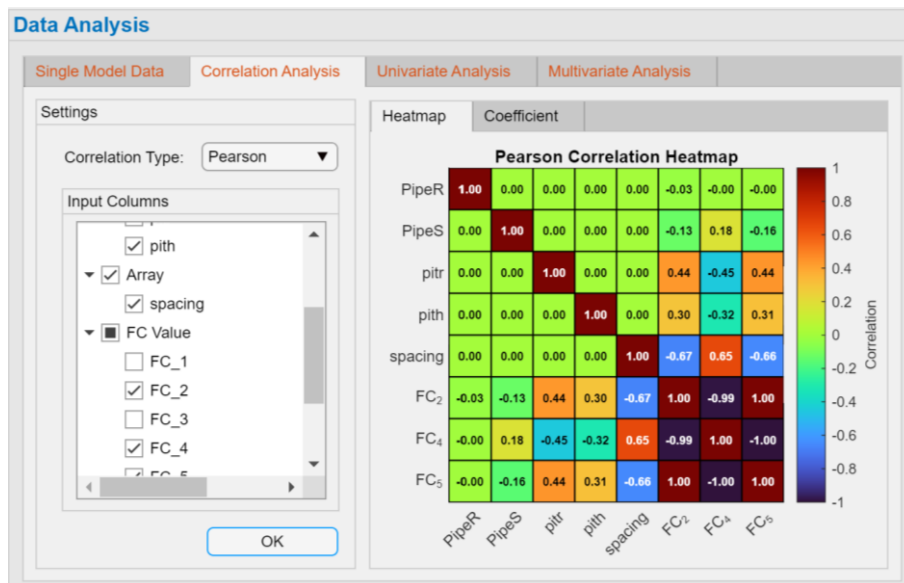
Select the target parameter combination via the dropdown box and click **OK**. The software displays the FC matrix in a table and visualizes its distribution with a 3D column chart.



(2) Correlation Analysis

The correlation analysis module quantifies the statistical correlation between input parameters and *FC* output values, identifying the dominant influencing parameters. Operation steps are as follows:

- Check the target variables in the **Input Columns** selection tree.
- Select a correlation algorithm (Pearson or Spearman) via the **Correlation Type** dropdown box.
- Click **OK** to run the analysis.
- Output results include a correlation matrix heatmap and a correlation coefficient table.



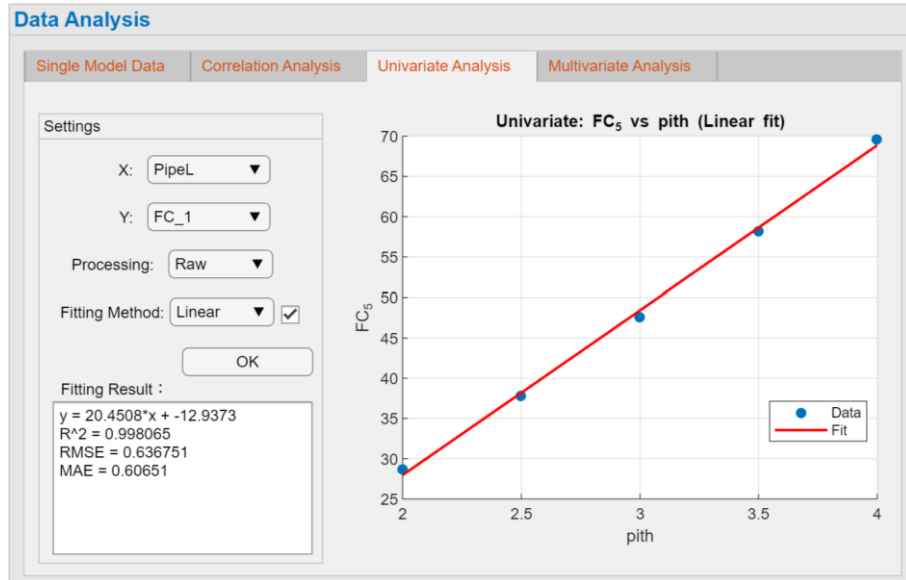
(3) Univariate Analysis

The univariate analysis module explores the quantitative relationship between a single input parameter and a single *FC* output value. The configurable parameters are as follows:

- **X**: Independent variable (single pipeline/pitting parameter).

- **Y:** Dependent variable (*FC* value).
- **Processing:** Data preprocessing modes (Raw / Logarithm / Reciprocal).
- **Enable Fitting:** Check to activate the curve fitting function.
- **Fitting Method:** Select Linear or Polynomial fitting.

The output includes a scatter plot, overlaid with a red fitting curve when fitting is enabled, together with quantitative evaluation results such as the fitting equation, R^2 , RMSE, and MAE.

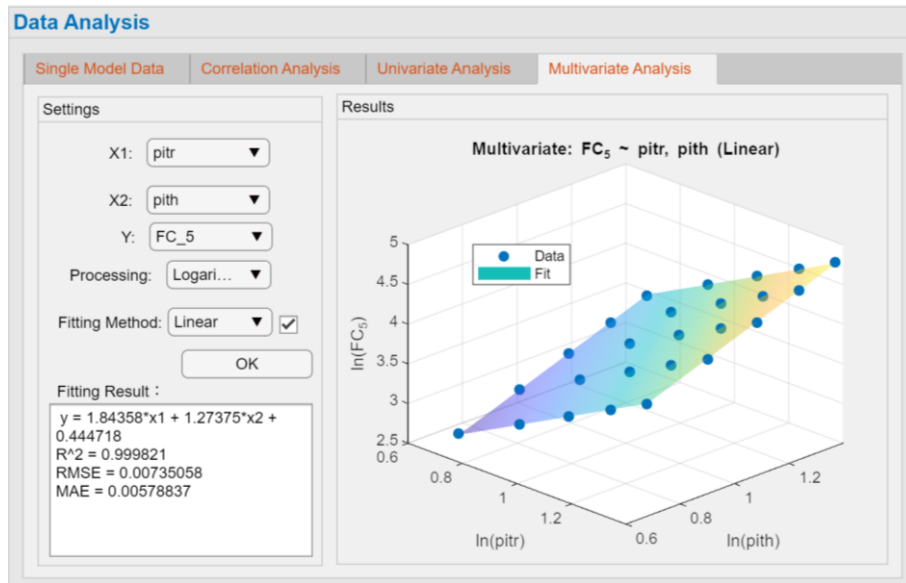


(4) Multivariate Analysis

As an extension of univariate analysis, this module supports joint regression analysis with two independent variables and one dependent variable. The configurable parameters are as follows:

- **X1, X2:** Two different independent variables.
- **Y:** Dependent variable (*FC* detection value).
- **Processing:** The same as in univariate analysis.
- **Enable Fitting:** Check to activate the surface fitting function.
- **Fitting Method:** Consistent with univariate analysis.

The output includes a 3D scatter plot. When fitting is enabled, a translucent fitting surface is overlaid, along with the complete regression equation and evaluation metrics (R^2 , RMSE, MAE).



3.4 Prediction Model Construction

This module provides a data-driven approach for estimating pitting corrosion parameters.

3.4.1 Data Preparation and Loading

Click **Load** to import Excel files with standard data column structures. The data import and filtering process is consistent with Section 3.3.1.

Select variables via the dual selection trees:

Input data (X): Model input variables.

Output data (Y): Model prediction target variables.

Load Data

Path: D:\data\train_data.xlsx Load

Input Data (X)

- FC Values
 - FC_1
 - ☒ FC_2
 - FC_3
 - ☒ FC_4
 - ☒ FC_5

Output Data (Y)

- PipeS
- ☒ Pits
 - ☒ pits
 - ☒ pith
- Array
- spacing

3.4.2 Model Selection and Hyperparameters

Select the required algorithm via the **Models** dropdown box, and adjust the corresponding hyperparameters in the **Settings** tab.

Model Select

Models: BP TrainRatio: 0.7 Train

Settings

BP

RF

GBDT

SER

epochs: 1000

goal: 0.0001

lr: 0.01

SER (Simultaneous Equation Regression) configuration steps:

- Set the number of regression equations (N) via the numeric spinner
- Select the dependent variable (Y) and two different independent variables (X1, X2) for each equation via the dropdown list
- Select the data processing mode (Raw / Logarithm / Reciprocal); an inverse transformation will be applied after coefficient estimation and prediction

Model Select

Models: SER TrainRatio: 0.7 Train

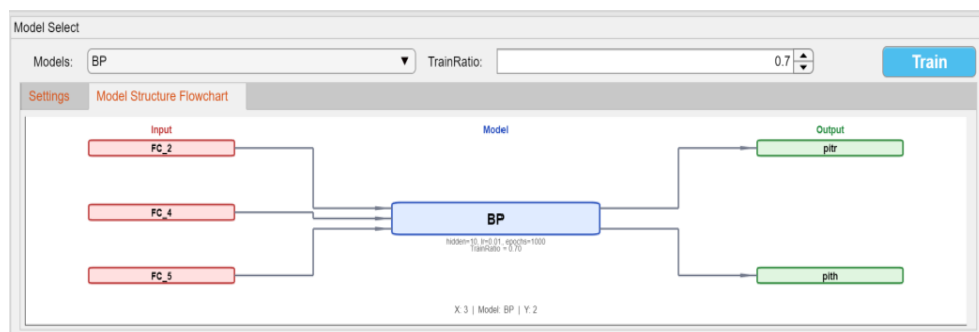
Settings Model Structure Flowchart

Processing: Raw Num: 2

Y	X1	X2
FC_2	ptr	pith
FC_5	ptr	pith

3.4.3 Training and Evaluation

Click the Train button to start model training. The real-time model structure flowchart will be displayed on the interface.

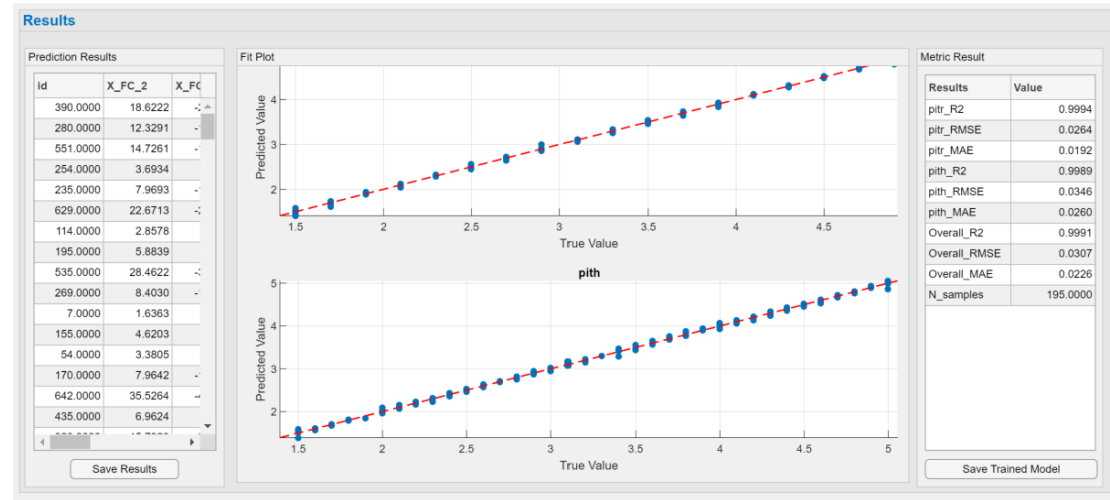


After training, the **Results** panel outputs the following contents:

- (1) **Prediction Results:** Input features, true values, and predicted values of test samples.
- (2) **Metric Results:** Quantitative evaluation metrics (R^2 , RMSE, MAE) of each output variable.
- (3) **Fit Plots:** Scatter plots of true values vs. predicted values, with the red dashed line representing the $y=x$ reference line.

Post-training operations:

- **Save Trained Model:** Save the structure (PredTrainedModel) as a .mat file.
- **Save Results:** Export prediction data and evaluation metrics as Excel files.



3.5 Model Prediction (Optional)

The saved prediction model can be used to predict pitting parameters. Relevant executable programs are stored in the `./prediction` directory, and detailed operation instructions are provided in the `README.md` file within this folder.